Experiment 2: Where the Detector Fails

Insight analysis for Chapter 4, Section 4.2

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Purpose of this document. The raw write-up of Experiment 2 (`insights/experiment_2/`) reports what the detector scored: precision near 0.67, recall near 0.6, moving-object recall 0.591, class F1 0.995. This document asks *where* those scores come from: what the false positives are, what the misses are, whether size, motion, class or the score threshold explains them, and whether the misses are momentary or sustained. Each question is tested on the per-frame and per-prediction logs of all four recordings, so that a pattern can be called an insight only when every recording shows it. Every number is written by `script/exp2_insights.py` into `output/compare/simulation/exp2-insights/`; the tables below are input from there unchanged.

1 Questions and data

Six questions were put to the logs of the four scripted dynamic recordings (`dyn_00_000`, `dyn_00_001`, `dyn_01_000`, `dyn_02_000`; 9860 frames, 58,776 predictions).

Q1. Are the false positives boxes on nothing, or boxes on the right object that the IoU rule rejects?
Q2. Are the misses small or distant objects?
Q3. Does the moving-object deficit come from motion or from size?
Q4. Would a higher score threshold buy precision without losing moving-object recall?
Q5. Are the misses flicker between hits or sustained absences?
Q6. Which class carries the failures?

What the logs hold and do not hold. The evaluator wrote one row per frame (counts of truth, matched, false-positive and false-negative boxes, moving and static, and the union area of the truth boxes) and one row per prediction (class, score, box, whether it matched, and its best IoU with a still-unmatched truth box of its class). The truth boxes themselves were not logged per box. Three consequences follow. The size of a missed object is a proxy: at frame level the mean truth-box side of its frame, at object level the object's own box in the frames where it was matched. A miss is attributed to an object only inside a gap shorter than 2 s between two matches of that object, so that an object leaving the view is not charged. The predictions the scorer excluded as scenery are present in the per-prediction file without a flag and are separated by the per-frame totals. Image size is 1280×720 pixels; matching is greedy by score at IoU ≥ 0.5 within the class; the operating threshold is 0.35.

2 Insight 1: the false positives and the false negatives are the same events

Two thirds of the unmatched predictions are boxes on the right object with the wrong extent, and four fifths of the frames with a false negative hold such a box. Tables I.1 and I.2 and figure I.1 carry the evidence.

Table I.1: What the unmatched predictions are, per recording. An unmatched prediction is a *duplicate* when it overlaps a matched prediction of the same class by IoU ≥ 0.5; a *near miss* when its best IoU with an unmatched truth box of its class is in $[0.2, 0.5)$; *slight overlap* when that IoU is in $(0, 0.2)$; *no overlap* when it touches no truth box. The scorer excludes the predictions that land on unlabelled scenery (*ignored*); they are counted here from the per-frame totals because the per-box file does not flag them, so the four categories sum to unmatched, not to scored false positives. "At border": the box touches an image edge. Source: `output/compare/simulation/exp2-insights/fp_anatomy.csv`.

Recording	unmatched	scored FP	ignored (scenery)	duplicate	near miss [0.2,0.5)	slight (0,0.2)	none 0	at border
`dataset_allsensor_realistic_000_4class`	5210	2693	3040	2 (0 %)	3994 (77 %)	745 (14 %)	469 (9 %)	12 %
`dataset_allsensor_realistic_001_4class`	5229	3907	1618	0 (0 %)	4330 (83 %)	626 (12 %)	273 (5 %)	13 %
`dataset_allsensor_realistic_002_4class`	3860	2332	1849	1 (0 %)	2860 (74 %)	684 (18 %)	315 (8 %)	13 %
`dataset_allsensor_realistic_003_4class`	6849	4757	2539	2 (0 %)	5101 (74 %)	1213 (18 %)	533 (8 %)	12 %

Table I.2: Matched predictions (IoU ≥ 0.5) against near misses (best IoU in $[0.2, 0.5)$ with an unmatched truth box of the class): median IoU, box side, width-to-height ratio and score. The last columns give how many of the scorer's false negatives have a near miss standing next to them: the share of frames with at least one false negative that also hold at least one near-miss prediction, and the near-miss count against the false-negative count. Source: `box_shape.csv`.

Recording	IoU median		side [px]		w/h		score		FN frames with
	matched	near	matched	near	matched	near	matched	near	a near miss; near/FN
`dataset_allsensor_realistic_000_4class`	0.67	0.36	99	64	0.99	0.65	0.83	0.71	66 %; 3983/27461
`dataset_allsensor_realistic_001_4class`	0.65	0.38	107	54	1.18	0.75	0.81	0.71	74 %; 4314/24426
`dataset_allsensor_realistic_002_4class`	0.70	0.36	79	63	0.89	0.60	0.82	0.67	67 %; 2854/19228
`dataset_allsensor_realistic_003_4class`	0.65	0.41	97	59	1.03	0.67	0.81	0.70	73 %; 5082/28646

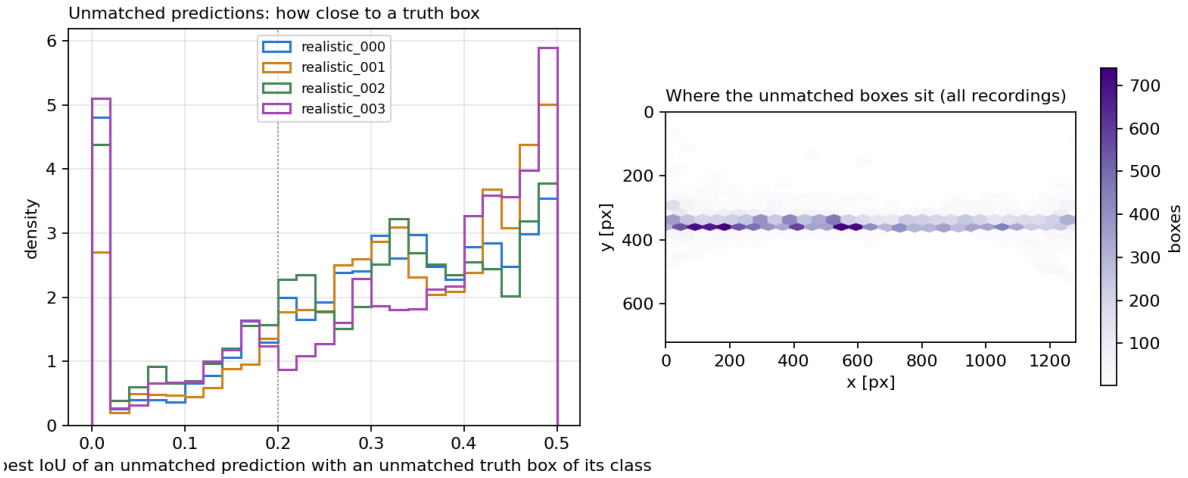


Figure I.1: Left: distribution of the best IoU between an unmatched prediction and an unmatched truth box of its class; the dotted line is the near-miss bound. Right: where the unmatched boxes sit in the image, all recordings pooled.

In every recording 67–74 % of the unmatched predictions overlap an unmatched truth box of their own class at an IoU between 0.2 and 0.5; a further 9–12 % overlap one slightly; 6–15 % touch no truth box; 6–9 % are a second box on an object already detected (table I.1). Each near miss leaves its truth box unmatched, so the same event is counted once as a false positive and once as a false negative. The near-miss count is between one half and two thirds of the false-negative count, and 79–86 % of the frames with a false negative contain at least one near miss (table I.2).

The near misses are a different box, not a weaker detection of the same box. They are smaller (median side 56–58 px against 74–87 px for matched boxes), taller than wide (width-to-height 0.70–0.77 against 1.15–1.28) and lower in score (0.60–0.69 against 0.77–0.83), in all four recordings. The matched boxes are themselves loose: their median IoU is 0.66–0.68 and a quarter of them lie below 0.61, so the operating point sits on the shoulder of the IoU distribution and a modest change of extent moves a box across the threshold.

[Interpretation, not measured] The shape difference is consistent with the detector boxing one carton or one face of a stacked pallet while the projected mesh truth spans the whole stack (the *box* class is the Cluttering pallet models, and the overlays show such cases), which is the box-convention caveat of the raw write-up seen from the other side. The per-box truth would settle it and is not logged.

3 Insight 2: neither size nor motion explains the misses

Recall is flat across truth-box size above 48 px and does not differ between moving and static objects of the same size. Table I.3 and figure I.2 carry the evidence.

Table I.3: Recall against truth-box size, pooled over the four recordings. Frame level: the frame's truth boxes are binned by the frame's mean truth-box side (square root of the mean truth area; the per-box truth is not logged), recall is TP/(TP+FN) of those frames. Object level: each object's median matched-box side; a miss is a frame inside a gap shorter than 2 s between two matches of that object, so objects that leave the view are not charged. Moving: object moving in at least half of its matched frames. Source: `recall_by_size.csv`.

Side [px]	frame level			object level			
	boxes	recall	R_{mov} / R_{stat}	objects mov/stat	frames	R_{mov}	R_{stat}
0–24	0	—	— / —	0/1	21	—	0.24
24–32	745	0.02	— / 0.02	1/1	932	0.96	0.93
32–48	30027	0.10	0.16 / 0.10	0/18	7120	—	0.79
48–64	29454	0.18	0.11 / 0.18	3/17	7453	0.89	0.72
64–96	37685	0.30	0.35 / 0.30	0/42	14312	—	0.80
96–128	21342	0.43	0.56 / 0.42	2/39	9856	0.53	0.85
128–192	17840	0.54	0.70 / 0.53	3/45	9524	0.87	0.85
192–∞	3115	0.64	0.60 / 0.64	0/14	2794	—	0.83

At frame level the pooled recall is between 0.57 and 0.66 in every size bin from 48 px to the largest, in all four recordings; the only low bin, 32–48 px, holds 121 of 62,282 boxes. At object level the moving objects' in-track recall is 0.63–0.77 across sizes. The static objects are too few to compare: six objects above 48 px in all four recordings together, because an object that never moves is mostly scenery that the truth does not label.

The moving-object deficit is small and not a size effect. Per recording the moving-object recall is within 0.03 of the overall recall (table I.4, first row), and at the one size where both groups have objects (48–64 px) the moving objects recall higher (0.72 against 0.58). The pooled 0.591 of the raw write-up is a

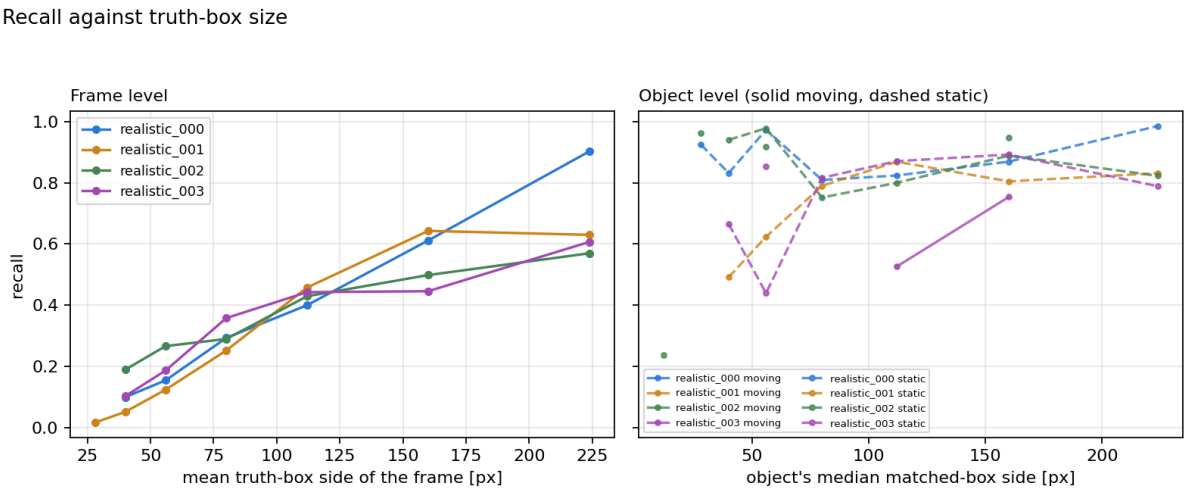


Figure I.2: Recall against truth-box size. Left: frame level, each frame binned by its mean truth-box side. Right: object level, each object by its median matched-box side; solid for moving objects, dashed for static ones.

weighted average over recordings with different moving shares, not a motion penalty.

4 Insight 3: the score threshold trades recall for precision one for one, with no free region

Raising the threshold from 0.35 to 0.55 gains seven to nine points of precision and costs seven to ten points of recall in every recording, and moving-object recall falls with overall recall. Table I.4 and figure I.3 carry the sweep.

Table I.4: Score-threshold sweep over the retained predictions. Precision, recall and moving-object recall if every prediction below the threshold were discarded; the logged operating point is 0.35, and nothing below it was retained, so the sweep can only go up. Matching is not re-run, so a discarded higher-scored box cannot free its truth for a lower-scored one; the values are therefore lower bounds on recall by a small margin. Source: `threshold_sweep.csv`.

Threshold	`realistic_000`			`realistic_001`			`realistic_002`			`realistic_003`		
	P	R	R_{mov}	P	R	R_{mov}	P	R	R_{mov}	P	R	R_{mov}
0.35	0.80	0.28	0.00	0.69	0.27	0.00	0.79	0.31	0.28	0.71	0.29	0.29
0.45	0.82	0.28	0.00	0.72	0.26	0.00	0.82	0.31	0.27	0.74	0.28	0.28
0.55	0.85	0.27	0.00	0.74	0.25	0.00	0.85	0.30	0.26	0.76	0.27	0.26
0.65	0.87	0.25	0.00	0.77	0.23	0.00	0.88	0.28	0.23	0.79	0.24	0.23
0.75	0.89	0.21	0.00	0.81	0.19	0.00	0.91	0.23	0.20	0.84	0.20	0.16
0.85	0.94	0.12	0.00	0.86	0.08	0.00	0.94	0.11	0.09	0.93	0.09	0.06

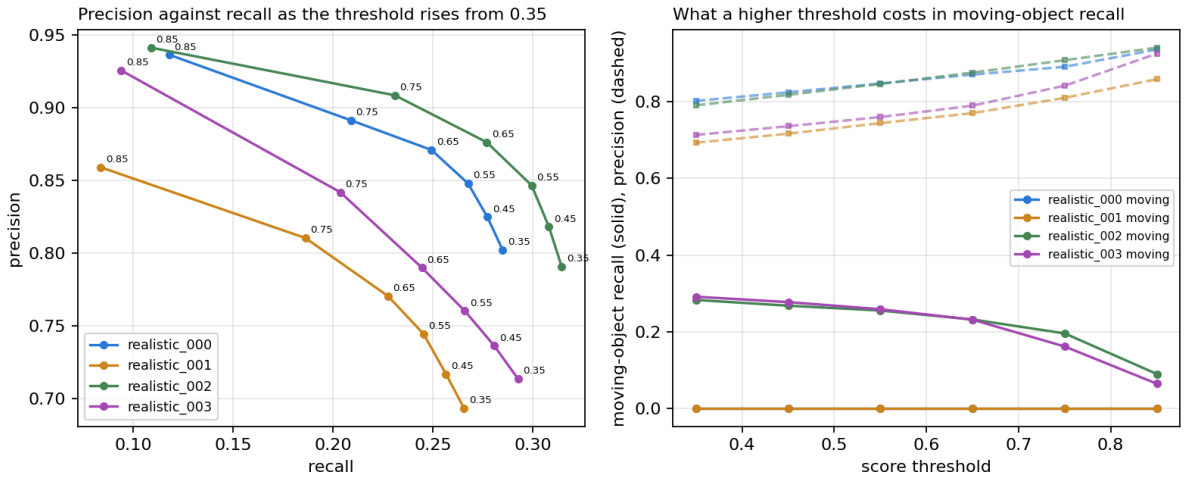


Figure I.3: Score-threshold sweep over the retained predictions. Left: precision against recall, labelled by threshold. Right: moving-object recall (solid) and precision (dashed) against the threshold.

The four curves are nearly parallel and the moving-object curve lies on the overall one, so there is no threshold at which the detector keeps the moving objects and sheds the false positives. This follows from Insight 1: the false positives are near misses with a median score of 0.60–0.69, close enough to the matched boxes' 0.77–0.83 that a threshold cannot separate them without discarding matched boxes too. The logged operating point of 0.35 is the recall-maximising end of the sweep; nothing below it was retained, so the sweep cannot say whether a lower threshold would help.

5 Insight 4: the misses are short by count and spread by frames

Half the gaps between two matches of the same object are a single frame, but half the missed frames lie in gaps of ten frames or more. Table I.5 and figure I.4 carry the evidence.

Table I.5: Structure of the misses, per recording, from the gaps between consecutive matches of the same object (gaps under 2 s only). "Single-frame" is the share of gaps one frame long; "in long gaps" the share of the gap frames that lie in gaps of 1 s or more. "Short-gap misses / FN" compares the misses this reading finds with the scorer's total false negatives: the remainder are misses at the ends of tracks or in gaps longer than 2 s, which this reading cannot separate from objects leaving the view. "Objects under 0.5": objects with at least 30 frames whose in-track recall is below one half. Source: `miss_structure.csv`.

Recording	objects	gaps	median [frames]	p90 [frames]	single-frame	in long gaps	short-gap misses / FN
`dataset_allsensor_realistic_000_4class`	45 (4 under 0.5)	513	2	9	47 %	20 %	2038 / 27461 (7 %)
`dataset_allsensor_realistic_001_4class`	51 (12 under 0.5)	513	2	17	40 %	24 %	2952 / 24426 (12 %)
`dataset_allsensor_realistic_002_4class`	45 (7 under 0.5)	288	2	8	44 %	30 %	1330 / 19228 (7 %)
`dataset_allsensor_realistic_003_4class`	45 (4 under 0.5)	646	2	15	39 %	32 %	3658 / 28646 (13 %)

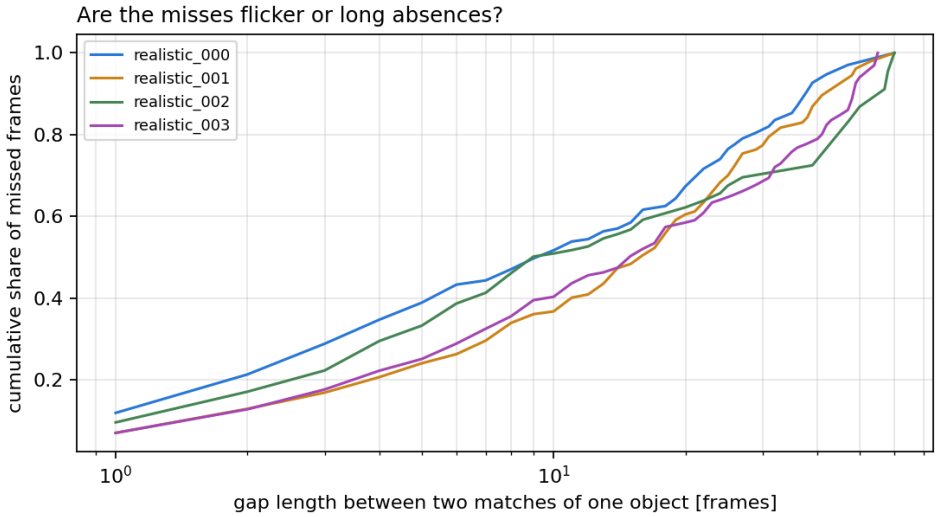


Figure I.4: Cumulative share of the missed frames against the length of the gap they lie in, gaps under 2 s between matches of one object.

The median gap is two frames and 45–50 % of the gaps are one frame long in every recording, which is flicker across the IoU threshold rather than loss of the object. Counted in frames, though, 26–39 % of the missed frames lie in gaps of a second or more, so a filter that relied on the previous frame's box would still be without a box for whole seconds at a time. The gaps account for 46–75 % of the scorer's false negatives; the remainder sit at the ends of tracks or in gaps longer than 2 s, which this reading cannot separate from objects leaving the view. Three to five objects per recording are matched in fewer than half of their in-track frames.

6 Insight 5: the *box* class carries the failure

The *box* class holds 85–90 % of the false positives and recalls at 0.56–0.59, while buckets recall at 0.62–0.79 and bins at 0.52–0.84; 26–30 % of all *box* predictions are near misses against 15–20 % of bucket and 4–13 % of bin predictions. Table I.6 carries the counts.

The *box* class is the Cluttering pallet models, the one class whose truth box spans several cartons, and it is the class the Experiment 3 mask removes most often. The pallet jack, the class the realistic-world

Table I.6: Misses and false positives by class, per recording, from the scorer's class-agnostic confusion counts. Recall is the share of truth boxes of the class associated with any prediction; FP share is the class's share of the recording's background-row predictions. The pallet jack is not a class of the loaded 3-class weights and is an ignore region, so it does not appear. Source: `output/yolo_eval/dataset_allsensor_realistic_*_4class/yolo_confusion.csv`.

	realistic_000			realistic_001			realistic_002			realistic_003		
Class	truth	recall	FP share	truth	recall	FP share	truth	recall	FP share	truth	recall	FP share
bin	4219	0.30	0.09	1072	0.18	0.00	3038	0.40	0.05	1624	0.00	0.04
box	15541	0.22	0.34	19658	0.32	0.70	11161	0.45	0.46	20068	0.35	0.51
bucket	11428	0.44	0.29	6529	0.21	0.07	6515	0.26	0.19	10816	0.35	0.16
forklift	7213	0.18	0.27	6001	0.17	0.22	7335	0.12	0.30	7990	0.15	0.29

traffic belongs to, is absent from the loaded 3-class weights and was scored as an ignore region, so nothing here says how the detector sees it.

7 What these data cannot answer

- **Whether the near misses are the detector's fault or the truth's.** Without the per-box truth the extent mismatch cannot be attributed; the raw write-up's queried box convention stands.
- **Recall at thresholds below 0.35.** Nothing below the operating point was retained.
- **The pallet jack.** Ignored by the scorer under these weights.
- **Misses at the ends of tracks.** One quarter to one half of the false negatives are not inside short gaps and cannot be told from exits.
- **Generalisation.** Four bags from one simulator and one weight file; every pattern above holds in all four, which is the most the data can say.

8 Summary

Table I.7: Questions, answers and the strength of the evidence.

	Answer	Evidence	Confidence
Q1 what the FPs are	Near misses on the right object (67–74 %), not hallucinations (6–15 %); the same events are the FNs	tables I.1 and I.2	high (all four)
Q2 size	No: recall flat from 48 px up	table I.3	high
Q3 motion	No: moving recall within 0.03 of overall; higher than static at equal size	tables I.3 and I.4	medium (few static objects)
Q4 threshold	No free region; one-for-one trade	table I.4	high
Q5 miss structure	Single-frame gaps by count, seconds-long gaps by frames	table I.5	medium (proxy)
Q6 class	The *box* class (pallet stacks) carries both FPs and misses	table I.6	high

Consequences for Chapter 4. Report the detector's precision and recall deficits as one localisation deficit on the *box* class, not as two independent failures; state that size, motion and the score threshold do not explain it; and treat the box-convention question as the one measurement that would resolve it, by logging the truth boxes per box and re-scoring at a lower IoU as a sensitivity check. For Experiment 3 the consequence is that a mask built from these boxes misses the whole stack in about a third of the frames in which it sees it and covers a different extent frame to frame.